Programming with AI – WOHI1

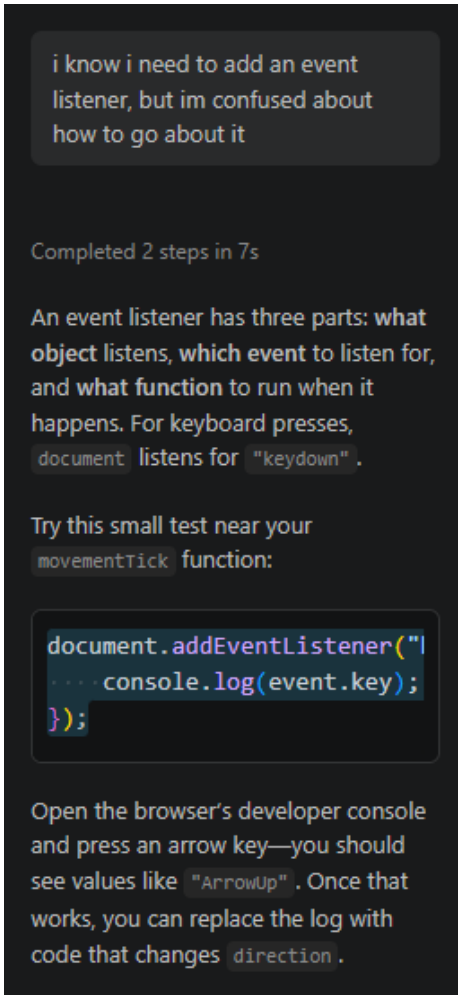
In this assignment I built a working Snake game using the provided starter file and with the help of a Copilot agent specifically instructed to act as a coding tutor to help me learn and make discoveries on my own, while also providing guidance whenever I would get stuck.

Part 1. Setup and agent configuration

I had GitHub Copilot set up already in Visual Studio Code and have used it occasionally when coding. I would already use it for learning before, but it didn't have any configuration. Setting up the custom agent was new to me, but the process was very easy, especially since after placing the agent file in my project I was able to ask Copilot to set it up for me.

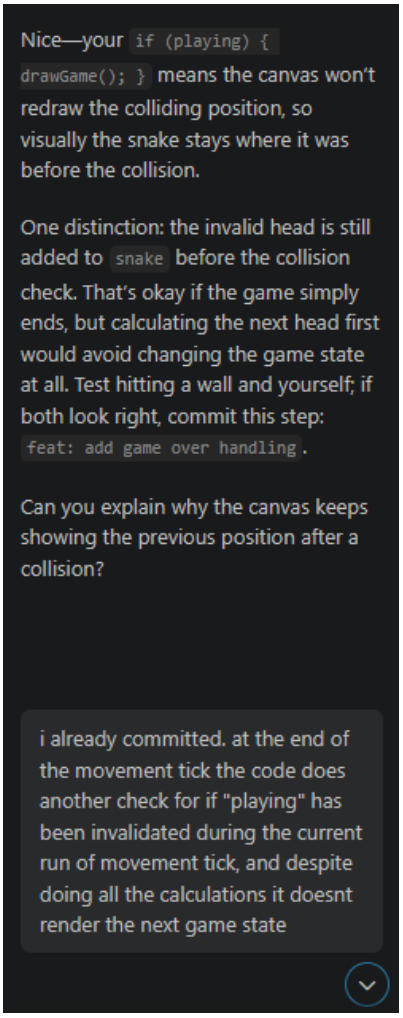
There were a few small changes I ended up making to the file so that the agent would act as neutrally as possible, removing some of the motivational fluff from it. I didn't have a frame of reference for what other kind of preferences I would have, so I kept the overall instructions mostly the same.

Part 2. Development process



This was the first instance of the agent proving to be more useful than an unconfigured AI. It was able to explain the EventListener at an appropriate complexity level.

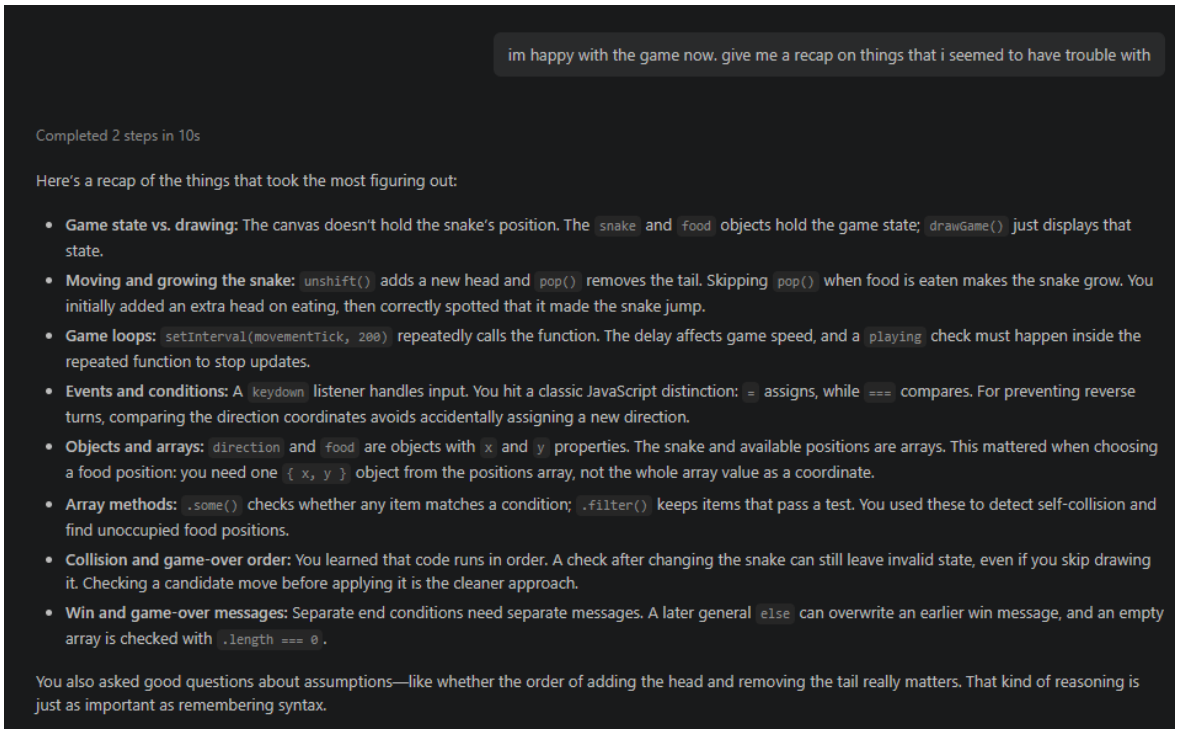
It was a bit surprising that it gave me the entire syntax, but it didn't give the correct code, just explained how to make it work, and then I was able to develop the feature on my own.



Two interesting things happened here. The agent struggled to keep up with my commits and kept reminding me to do so even if I had already done it. Seems like an oversight.

The occasional questions asking to explain a feature of the code were nice. Taking a quick pause to reflect over my code and make sure I understand all of it was productive.

Recap at the end was an excellent way to end things off. It was nice to go through newly learned concept and ideas. However, it's important to still be critical towards the agents reasoning. I did not struggle with the concept of code running for example, that is a very basic principle. I ended up adding a few more features after this still, so that it was a bit more complete.



Part 3. Reflection

The agent configuration made Copilot more suitable for learning, as opposed to just results oriented programming. Basic concepts like EventListeners, Arrays and Objects were basically mandatory knowledge for this, and the lectures and practice sessions made development of the game feel quite easy. I had the inline code suggestions on a massive delay, so they barely showed up at all, but they of course don't care about the learning aspect of it and simply complete the code.

Some concepts such as a few Array related Methods were unfamiliar to me, as well as the usage of setInterval, but overall, I didn't really end up needing to learn too many complex or completely unfamiliar concepts for this project. Everything was just an extension or application of something I was already familiar with.

Maybe there exists some smart ideas for what to put in the agent configuration, but for me this kind of tutor was essentially what I was trying to achieve. The agent might need more context and time to slowly learn what I'm struggling with to set up the most effective learning moments. When the agent suggested something unhelpful, I just moved past it with another specifying question. The suggestions were generally decent, so there wasn't a need to take more serious action.

I don't think having a specifically configured agent is different from just using Copilot on the control-automation axis. The default configuration has its own parameters for what to help with and is already able to explain and teach if asked to, or if it determines the user to not be particularly advanced at coding. Both count as AI-powered IDEs, as they can affect the environment in more ways than just inline edits.

Finally, here is a list of my additional features besides the required functionality:

- Input buffer system for more responsive gameplay
- Can't turn 180 degrees and instantly die
- Game over and You win screens (Fill the board in its entirety to win, good luck :D)
- Play again button
- Score counter
- Centered the game on the screen
- Game gets faster as it progresses